

**From Research Question to Sample: A
Decision Framework for Research Design,
Sampling Technique, and Sample Size in
Senior High School Studies**

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ABSTRACT

This article develops a practical planning framework from the research-design and sampling material presented in *Statistics Made Easy*. It organizes the source book's descriptions of quantitative research designs, probability and non-probability sampling techniques, sampling bias, and sample-size estimation into a sequence of decisions that senior high school researchers can apply before data collection. The framework begins with the purpose of the research question and distinguishes descriptive, correlational, causal-comparative, quasi-experimental, true-experimental, survey, pre-experimental, and evaluation designs as presented in the source. It then connects the accessibility and structure of the target population to simple random, systematic, stratified, cluster, convenience, quota, purposive, and snowball sampling. The article further explains the role of sampling bias and summarizes the source book's use of Cochran's and

Slovin's formulas for sample-size estimation. Rather than treating sample size as an isolated computation, the article places it after population definition and sampling-design selection. The proposed framework is intended to help student-researchers document why a particular design and sampling strategy was selected, what limitations it creates, and how the resulting sample supports the research question.

Keywords: *research design; sampling technique; sample size; Cochran formula; Slovin formula; sampling bias; senior high school*

INTRODUCTION

A research study can produce a correctly computed statistic and still be weak if the design and sample do not support the question being asked. Statistics Made Easy places research design and sampling before the chapters on statistical analysis, reflecting an important planning principle: the way respondents or observations are selected determines what the resulting data can reasonably represent.

This article reorganizes the source material into a decision framework for senior high school researchers. The objective is not to replace the definitions in the book but to show how they connect. The central sequence is research purpose -> design -> target population -> sampling strategy -> sample size -> bias control -> documentation.

Selecting a Quantitative Research Design

The source lists several quantitative designs. Descriptive research is used when the purpose is to provide an

accurate and systematic description of a population, situation, or phenomenon. Correlational research measures variables and examines their statistical relationship without experimental manipulation. Causal-comparative research compares groups in an attempt to examine the effect of an independent variable on a dependent variable. Quasi-experimental research resembles experimentation but does not fully meet the conditions of a true experiment. True-experimental research is associated in the source with random assignment. Survey research centers on collecting data from respondents. Pre-experimental designs expose participants to an intervention without the comparison structure described for stronger experimental designs. Evaluation research is presented as a way to obtain constructive feedback that can support decisions or policy.

The educational value of these categories is greatest when students connect them directly to the verbs in the research question. Questions asking what exists or to what extent point toward description. Questions asking whether two variables are related point toward correlation. Questions asking whether groups differ or whether an intervention has an effect require a design capable of supporting that comparison.

Design-to-Question Planning Matrix

Design	in the source	Typical purpose	Illustrative question form	Key planning feature
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Descriptive	Describe a population, condition, or phenomenon			
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What is the level/status/profile of ...?

Is there a relationship between X and Y?

Two measured variables; no experimental manipulation
Clear variables and descriptive measures Correlational
Examine statistical relationship between variables
Causal-comparative Compare existing groups Is there a
difference between groups associated with X?
Does an intervention change an outcome?
Pre-existing groups and a measured outcome Quasi-
experimental Estimate an intervention effect without full
true- experimental conditions
Comparison structure without complete random
assignment True-experimental Test an intervention
under experimental conditions
Random assignment emphasized in the source
Does treatment A cause a different outcome than
control?
Survey Collect respondent data What do respondents
report about ...?

Limited comparison structure

What happens after the intervention?
Questionnaire or structured response collection Pre-
experimental Observe an intervention without the
source's stated control/experimental comparison
Evaluation Gather feedback for decisions or policy
How is a program, service, or policy evaluated?

Decision-oriented information

Probability Sampling as Structured Selection

The book presents simple random, systematic, stratified,
and cluster sampling as probability techniques. Simple
random sampling gives each population member an
equal chance of selection. Systematic sampling selects
members at a fixed interval after establishing the

sampling frame. Stratified sampling divides the population into subgroups and ensures representation from relevant strata. Cluster sampling selects groups or clusters rather than selecting every individual directly.

A student-researcher can choose among these approaches by examining how the population is organized. If a complete list exists and individual selection is feasible, simple random or systematic selection may be practical. If meaningful subgroups must be represented, stratification becomes useful. If the population is naturally organized into intact groups, cluster sampling may reduce the logistical burden of individual selection.

Non-Probability Sampling and Access Constraints

The source also presents convenience, quota, purposive, and snowball sampling. Convenience sampling relies on availability and willingness. Quota sampling seeks specified numbers of participants from defined categories. Purposive sampling uses researcher judgment to select participants with relevant characteristics. Snowball sampling grows through participant referrals and is associated in the source with hard-to-reach populations.

These methods are often chosen because access is limited, but the reason for selection should be reported clearly. A non-probability sample should not be described as random if it was recruited through availability, researcher judgment, quotas, or referrals. Transparent labeling is part of methodological quality.

Sampling Bias as a Planning Risk

Statistics Made Easy identifies several sources of sampling bias: violating previously agreed sampling rules, excluding difficult-to-reach groups, replacing selected individuals, obtaining low response rates, and relying on an outdated sampling frame. These examples can be translated into a simple monitoring question: did the actual recruitment process depart from the planned sampling procedure in a way that changed who had a chance to participate?

A sampling log is therefore useful even in small student studies. The log can record the planned sample, selected participants, nonresponses, replacements, exclusions, and reasons for any deviation. This documentation does not eliminate bias, but it prevents the limitation from becoming invisible.

Determining Sample Size

The source presents two formulas. Cochran's approach is described as useful for large populations and is based on desired precision, confidence, and an estimated population proportion. A finite-population correction is then shown for smaller populations. Slovin's formula is presented as a simpler method that uses population size and a desired margin of error.

The key planning lesson is that a formula should be applied only after the target population has been defined. Sample-size computation cannot compensate for a poorly defined population or an inappropriate sampling procedure. The researcher should document the population size used, the margin of error or precision assumption, any estimated proportion required by the formula, and the final target sample.

Proposed Seven-Step Sampling Decision Framework

Step 1 - State the research question and the intended type of conclusion.

Step 2 - Choose the research design that best matches that conclusion.

Step 3 - Define the target population and identify whether a usable sampling frame exists.

Step 4 - Decide whether probability selection is feasible and appropriate. If not, name the non-probability method honestly.

Step 5 - Determine the sample size using the formula or rule adopted in the study and document every assumption used in the computation.

Step 6 - Implement the recruitment procedure without unrecorded substitutions or exclusions.

Step 7 - Report deviations, nonresponse, and other sampling limitations in the methodology and limitations sections.

Implications for Classroom Research

The source material is particularly useful for teaching because it presents both design labels and concrete sampling procedures. The proposed framework turns those topics into a sequence of decisions that can be reviewed by an adviser before data collection starts. Students can be required to submit a one-page sampling plan showing the population, design, sampling method, sample-size computation, and anticipated sources of bias.

This approach also helps separate two questions that are often confused: how many participants are needed and how those participants will be selected. Sample size

addresses quantity; sampling design addresses selection. Both are necessary, and neither substitutes for the other.

CONCLUSION

Research design, sampling technique, and sample size should be treated as an integrated planning problem. *Statistics Made Easy* provides the component concepts; this article reorganizes them into a framework that begins with the research question and ends with transparent documentation of the sample actually obtained. For senior high school researchers, the most important practical lesson is that the credibility of statistical analysis depends on choices made before computation begins.

REFERENCE

Simpal, M. T., & Mangakoy, A. B. (2022). *Statistics made easy: Commonly used statistical tools and its computer applications in quantitative research for senior high school (1st ed.)*. Shirley Publishing House.